

概率统计 21-22-2(A)标准答案及评分标准

一、选择题

1)C 2)D 3)C 4)B, 5)D

二、填空题

1)10./13=0.769;

2)0.0064

3)32

4)0.8413

5)1.5

6) -1/15=-0.067

7)1

8) 1/4=0.25

$$9) \quad F(x) = \begin{cases} 0 & x < -2 \\ 0.5 & -2 \leq x < 0 \\ 0.7 & 0 \leq x < 2 \\ 1 & x \geq 2 \end{cases}$$

$$10) \quad f_Y(y) = \begin{cases} 0.625 & 0 < y < 1 \\ 0.375 & 1 \leq y < 2 \\ 0 & \text{其它} \end{cases}$$

11) 0.5

12) [14.02, 15/98]

13) $\frac{6}{11} = 0.545$

三、

$$\text{三、} \quad (1) \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$$

$$a \int_0^1 \int_{-\sqrt{y}}^{\sqrt{y}} x^2 y dy dx = 1;$$

$$a = \frac{21}{4} \dots\dots\dots$$

$$(2) f_X(y) = \int_{-\infty}^{\infty} f(x, y) dy \dots$$

$$\text{当 } 0 < |x| < 1 \text{ 时 } f_X(x) = \int_{x^2}^1 ax^2 y dy = \frac{21}{8} x^2 (1 - x^4)$$

$$\text{当 } |x| \geq 1 \text{ 时 } f_X(x) = 0.$$

$$(3) f_{Y|X}(y|x) = \frac{f(x, y)}{f_X(x)} = \begin{cases} \frac{2y}{1-x^4} & x^2 < y < 1 \\ 0 & \text{其它} \end{cases} (|x| < 1).$$

$$f_{Y|X}(y|0.5) = \begin{cases} \frac{32}{15} y & \frac{1}{4} < y < 1 \\ 0 & \text{其它} \end{cases}$$

$$P(Y < 0.5 | X = 0.5) = \int_{0.5}^{+\infty} f_{Y|X}(y|0.5) dy = \int_{0.5}^1 \frac{32}{15} y dy = \frac{4}{5} = 0.8.$$

四、A 表示原来盒子中为白球；B 表示抽到白球，则

$$P(A) = P(\bar{A}) = \frac{1}{2};$$

$$P(B|A) = 1; P(B|\bar{A}) = 0.5;$$

$$\begin{aligned}(1) P(B) &= P(A)P(B|A) + P(\bar{A})P(B|\bar{A}) \\ &= 0.5 * 1 + 0.5 * 0.5 = 0.75\end{aligned}$$

(2)

$$\begin{aligned}P(A|B) &= \frac{P(A|B)}{P(B)} = \frac{P(A)P(B|A)}{P(B)} \\ &= \frac{0.5 * 1}{0.75} = \frac{2}{3} \approx 0.667.\end{aligned}$$

$$\begin{aligned}\text{五、 } Z \text{ 的分布函数 } F_Z(z) &= P(Z \leq z) = P(\max(X, Y) \leq z) \\ &= P(X \leq z, Y \leq z) ..\end{aligned}$$

当 $z < 0$ 时, $F_Z(z) = 0$;

$$\text{当 } 0 \leq z \leq \frac{1}{2} \text{ 时, } F_Z(z) = \iint_{x \leq z, y \leq z} f(x, y) dx dy .$$

$$\begin{aligned}&= 2 \int_0^z \int_0^z dx dy \\ &= 2z^2\end{aligned}$$

当 $\frac{1}{2} < z \leq 1$ 时

$$\begin{aligned}F_Z(z) &= \iint_{x \leq z, y \leq z} f(x, y) dx dy = 1 - P(X > z \cup Y > z) = 1 - 2 * 2 * \frac{1}{2} (1 - z)^2 \\ &= 1 - 2(1 - z)^2 .\end{aligned}$$

当 $z > 1$ 时, $F_Z(z) = 1$

Z 的概率密度为

$$f_Z(z) = [F_Z(z)]' = \begin{cases} 4z & 0 < z \leq \frac{1}{2} \\ 4(1 - z) & \frac{1}{2} < z < 1 \\ 0 & \text{其它} \end{cases}$$

六、X 表示 100 件产品中检测出的一等品个数, $X \sim b(n, p)$, $n = 100, p = 0.6$.

$$\mu = EX = np = 60; \sigma^2 = DX = np(1 - p) = 24;$$

所求概率为:

$$P(X > 68) \approx 1 - \Phi\left(\frac{68 - 60}{\sqrt{24}}\right).$$

$$= 1 - \Phi\left(\frac{4}{\sqrt{6}}\right) = 1 - \Phi(1.633)$$

七、(1)似然函数为: $L(\theta) = \prod_{i=1}^n f(X_i, \theta) = \prod_{i=1}^n 5e^{5(X_i - \theta)}$
 $= 5^n e^{5\sum_{i=1}^n X_i - 5n\theta}, \theta \geq \max(X_1, \dots, X_n).$

$L(\theta)$ 是 θ 的单调减函数, 所以, 当 $\theta = \max(X_1, \dots, X_n)$ 时, $L(\theta)$ 取最大值。

故 θ 的最大似然估计量为:

$$\hat{\theta} = \max(X_1, \dots, X_n).$$

(2) 总体的分布函数为:

$$F(x) =$$

$$\begin{cases} e^{5(x-\theta)} & x \leq \theta \\ 1 & x > \theta \end{cases}$$

$\hat{\theta}$ 的概密度函数为, $F_{\hat{\theta}}(t) = n[F(t)]^{n-1}f(t)$

$$= \begin{cases} 5ne^{5n(t-\theta)} & t \leq \theta \\ 0 & t > \theta \end{cases}$$

$$E\hat{\theta} = \int_{-\infty}^{\theta} t 5ne^{5n(t-\theta)} dt = \theta - \frac{1}{5n}$$

$$E\hat{\theta} \neq \theta, \hat{\theta}$$

不是 θ 的无偏估计

八、(1) $n = 25, \alpha = 0.05,$

$$\text{检验统计量 } T = \frac{\bar{X} - 24}{S_n} \sqrt{n} | H_0 \sim t(n-1)$$

拒绝域: $D = \{T > t_{\alpha}(n-1)\} = \{T > 1.7113\}.$

$$\bar{x} = 26, s_n = 4$$

$$T \text{ 的观测值: } T = \frac{26 - 24}{4} \sqrt{25} = 2.5$$

$$2.5 > 1.7113$$

所以, 拒绝原假设。

$$(2)\sigma^2 \text{ 的置信度为 } 95\% \text{ 的置信区间为: } \left[\frac{(n-1)S_n^2}{\chi_{0.025}^2(24)}, \frac{(n-1)S_n^2}{\chi_{0.975}^2(24)} \right]. \dots$$

$$= \left[\frac{24 \times 16}{39.36}, \frac{24 \times 16}{12.4} \right] = [9.756, 30.968]$$